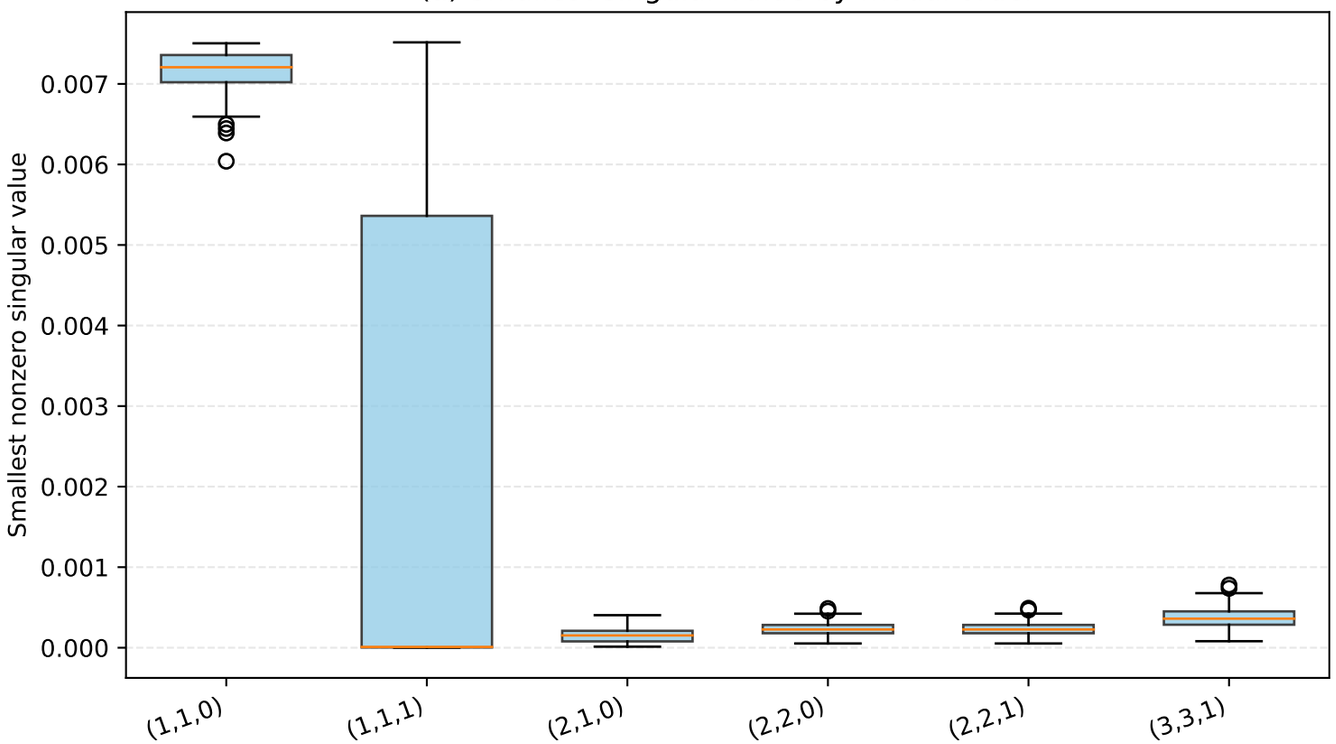
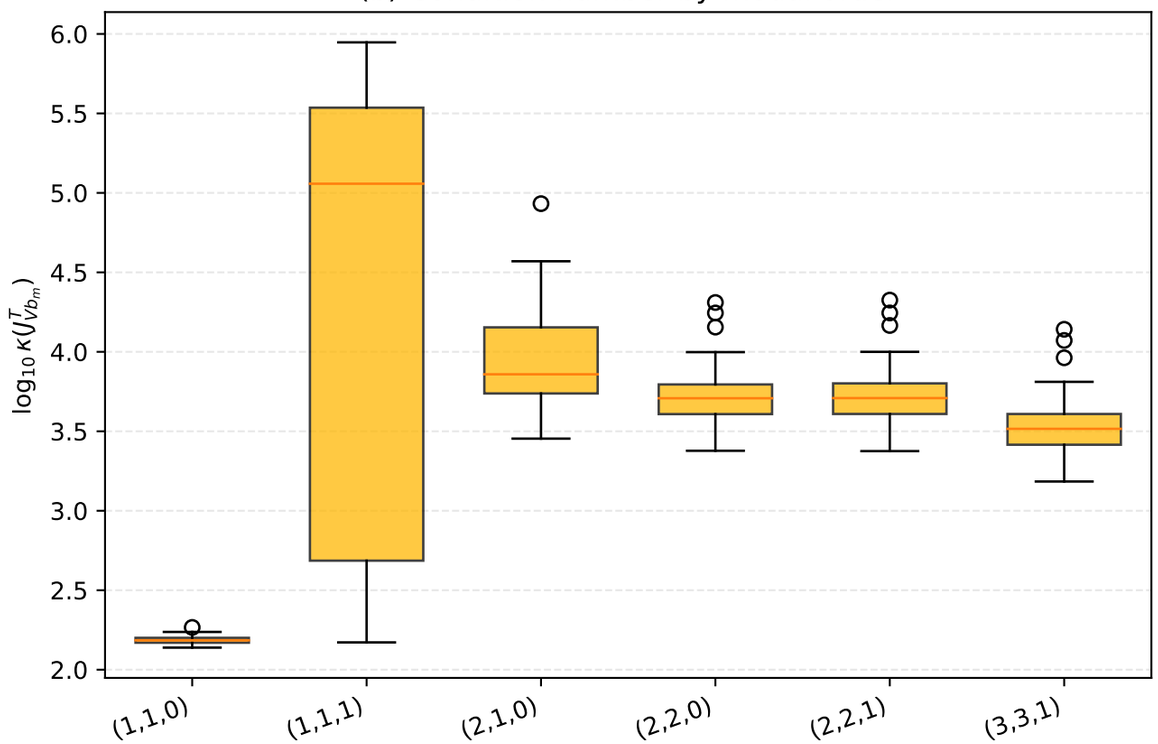
Tip-wrench observability vs modal order

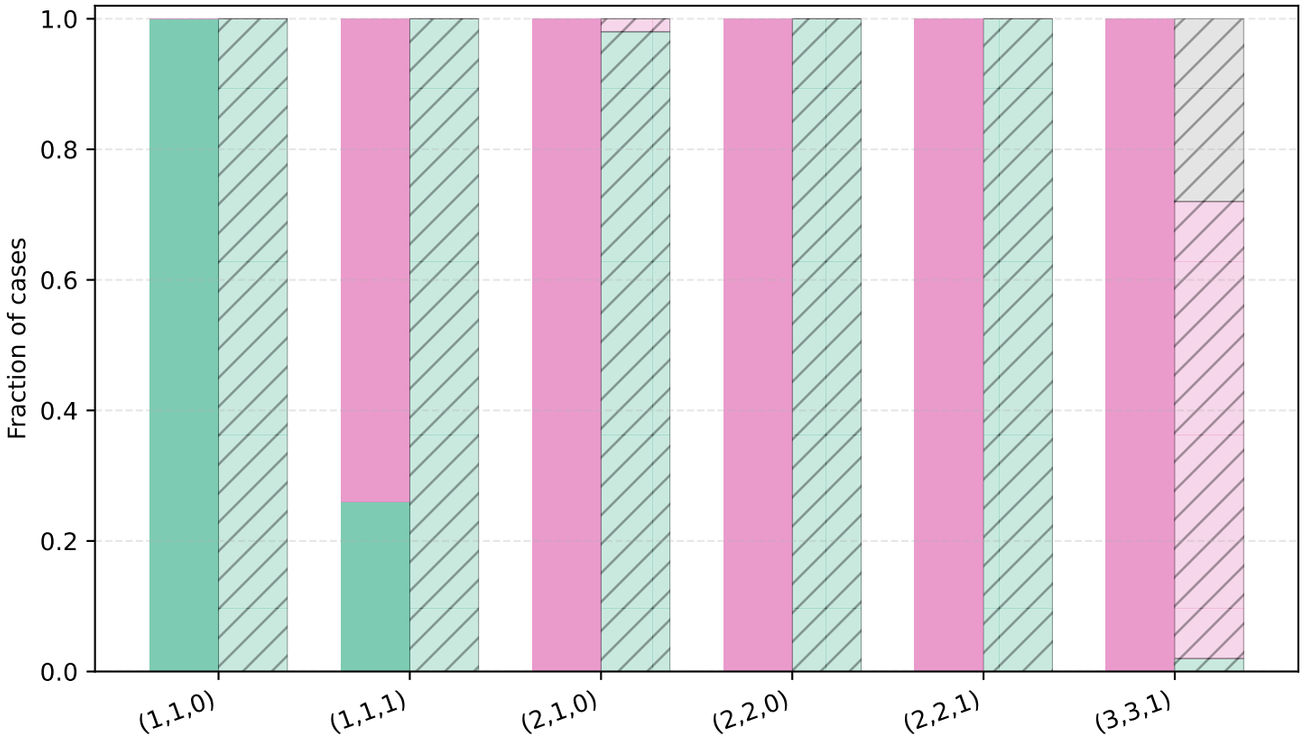
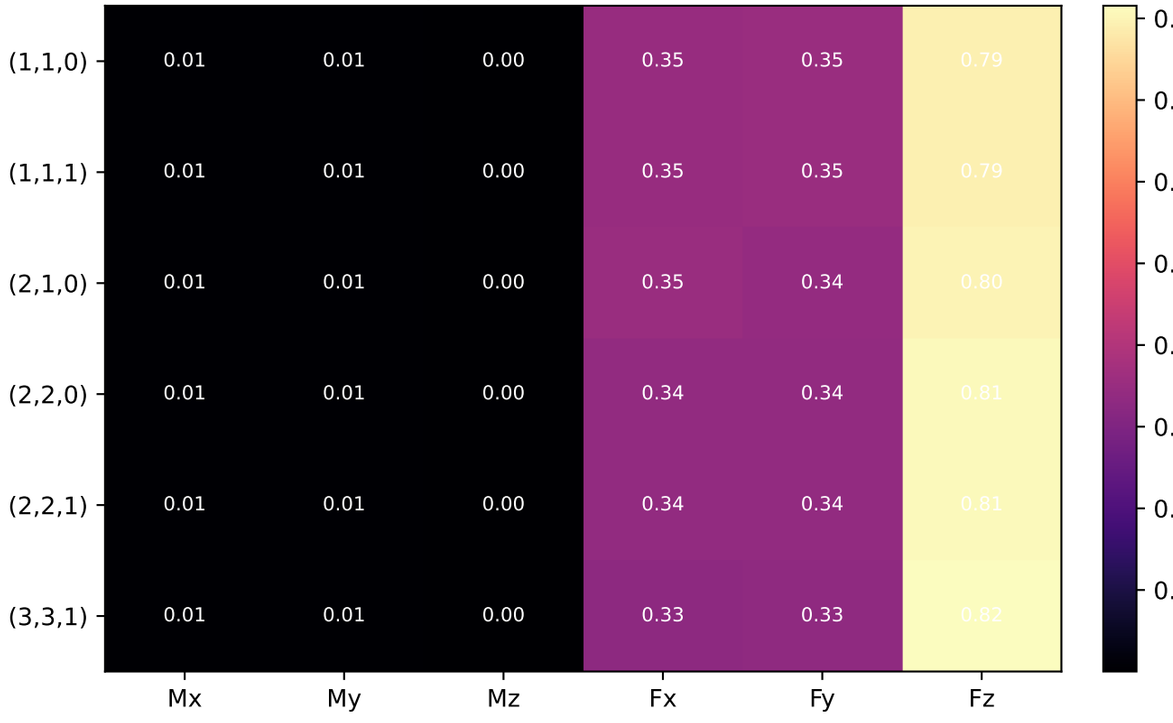
(a) Smallest singular value by modal order



(b) Condition number by modal order



(c) Numerical rank vs effective rank



(d) Avg. |weakest right singular vector|